

Automation cut audit data preparation time by 25%

n8n workflows and a Power BI model with rule-based exception flagging for forensic and controls engagements.

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Client names and identifying details have been withheld. Prepared by Archish Chugh for portfolio purposes; this is not an EY publication and does not represent the views of any client.

Automation cut audit data preparation time by 25%

Engagement teams were spending a large share of fieldwork budget collecting, cleaning and reconciling client data. n8n workflows and a Power BI model now handle most of that preparation, releasing time for control testing.

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USED ON

EY Risk & Compliance engagements

ROLE

Designed and built the workflows

TOOLS

n8n · Power BI · Excel

RESULT

25% faster data preparation

EXECUTIVE SUMMARY

SITUATION

Forensic and controls engagements received client data in many formats: PDFs, ERP exports, registers and spreadsheets.

COMPLICATION

Manual collection and matching slowed every engagement and reduced the time available for testing.

APPROACH

n8n workflows for ingestion and reconciliation, rule-based exception flags, and a Power BI star-schema model.

IMPACT

25% faster data preparation, used on 2–3 engagements, with full-population testing where the data allows.

25%

faster data pull and reconciliation

2–3

engagements using the workflows so far

100%

of transactions tested, instead of samples, where the data allows

01 · THE PROBLEM

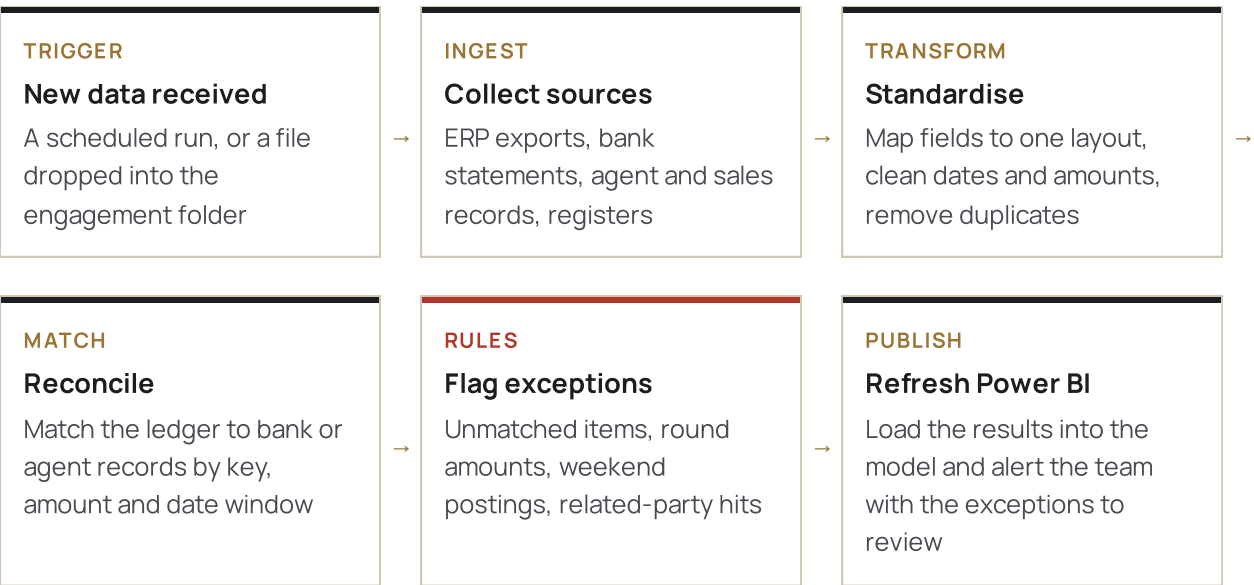
Manual data preparation was consuming fieldwork budget

On forensic and controls engagements, data arrives in many forms: bank statements as PDFs, ERP exports, agent sales files, collection registers and emailed spreadsheets. Analysts spent days collecting, reformatting and matching it by hand. The work was slow, error-prone and different on every engagement, and it cut into the time left for control testing.

02 · WORKFLOW

An n8n pipeline automates collection, matching and exception flagging

Exhibit 1: Reconciliation workflow, from source to exception list Representative design.



Here is how it worked on the aviation engagements. The workflow matched counter ticket sales against agent records and matched cash collections against the books. The exceptions it produced, such as collections never recorded and excess baggage not charged, became the starting point for control testing instead of something the team found at the end.

03 · TRY IT

Interactive demo: exception rules applied to a sample reconciliation

This is a small version of the matching and exception rules from the workflow. It uses sample data, not client records. Turn rules on or off to see how the exception list changes.

Exhibit 2: Collections, books against bank (sample)

Sample data for demonstration

9 items tested 3 matched 6 exceptions

DATE	REFERENCE	BOOKS ₹	BANK ₹	RESULT
Mon, 03 Mar	Agent A collection	48,250	48,250	Matched
Tue, 04 Mar	Agent B collection	1,20,000	1,20,000	Round amount
Wed, 05 Mar	Counter sales	36,780	35,780	Differs by ₹1,000
Sat, 08 Mar	Voucher V-114	25,500	25,500	Weekend posting
Mon, 10 Mar	Agent C collection	62,410	—	Not in bank
Tue, 11 Mar	Agent A collection	51,090	51,090	Matched
Wed, 12 Mar	Refund R-22	50,000	50,000	Round amount
Thu, 13 Mar	Unidentified receipt	—	18,600	Not in books
Fri, 14 Mar	Agent D collection	44,375	44,375	Matched

04 · DATA MODEL

A star-schema model links every exception to the control that should have caught it

Exhibit 3: Star schema behind the dashboards

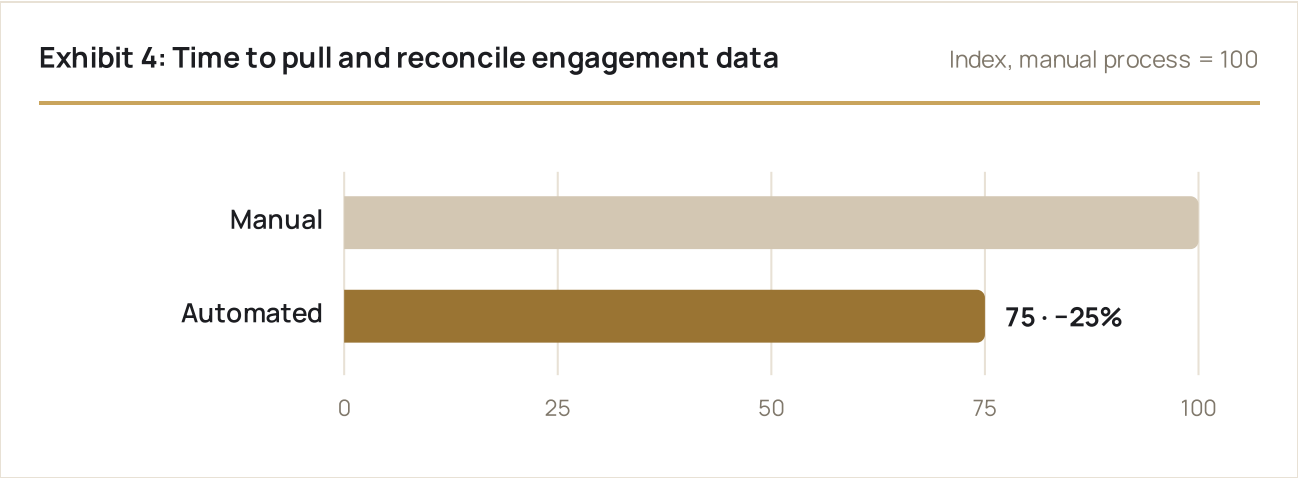
Representative.

Dim_Entity Legal entities, directors, shared addresses	Fact_Transactions Every ledger and bank line, with a match status and exception flags	Dim_Counterparty Vendors, agents and customers, with related-party tags
Dim_Date Period, fiscal year, weekend and holiday flags	Fact_Exceptions Rule hits, reviewer, disposition and evidence reference	Dim_Control Control ID, owner, COSO component, test result

The dashboards had three views. An engagement overview showed reconciliation coverage and open exceptions. An exception register let reviewers filter by rule, entity and value and record what they decided. A controls view linked each exception to the control that should have prevented it, so the audit report could go straight from the data to the root cause.

05 · RESULTS

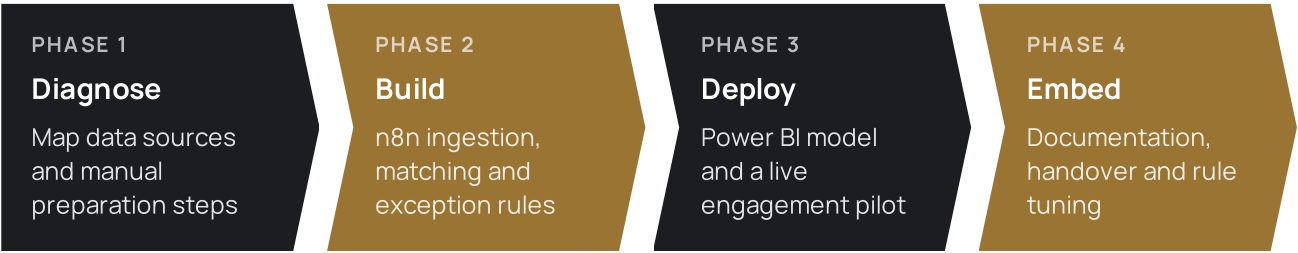
Data preparation time fell by 25%, and testing moved from samples to full populations



- **Productivity:** preparing engagement data took 25% less time, and that time went into control testing.
- **Assurance:** reconciliations cover full populations, so we test every transaction instead of a sample and exceptions can't fall between samples.
- **Consistency:** the same rules run on every engagement, and each exception links to its evidence reference for review.

06 · ADOPTION

Adoption was driven through a live pilot and a documented handover



- 1 Start with one real engagement**
Built the first workflow on a live engagement, so the time saved was measured on real work.
- 2 Show managers the exceptions**
Walked managers through the exception register and how each flag ties back to a control finding.

3**Write it down**

Documented each workflow's inputs, rules and outputs so other analysts can reuse it independently.

4**Improve from feedback**

Added and adjusted rules based on reviewers' decisions, which reduced false positives over time.

KEY TAKEAWAYS

- 1 Automating data preparation moves budget from collecting data to testing controls.
 - 2 Rules run on full populations catch exceptions that sampling misses.
 - 3 Adoption depends on a live pilot and documentation others can use.
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RELEVANCE TO UAE ROLES**Relevance to the UAE market**

UAE internal audit functions in banking, aviation and government are investing in continuous auditing and data analytics, following the IIA's Global Internal Audit Standards. This experience transfers directly to building continuous-auditing capability on whichever low-code or BI tools a team already runs.